Protection Without a Theory of Change: Tariff Policy and Import Substitution Outcomes in Nigeria, 2023-2025

A comparative case study of refined petroleum, automobiles, and rice

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Abstract

Nigeria's tariff regime is often read as a single coherent policy stance — protectionist, liberalizing, or caught between the two under its African Continental Free Trade Area (AfCFTA) commitments. This paper argues that framing is empirically inadequate. Using a comparative case-study design across three sectors subject to materially different tariff treatment between 2023 and 2025 — refined petroleum (an effective tariff near zero), automobiles (a 35-40% wall with a 0-10% concession for local assembly), and rice (a 70% combined duty and levy, among the highest in Nigeria's schedule) — this paper finds that effective tariff level does not predict import-substitution outcomes. Petroleum, the least-protected sector, recorded the fastest and most durable substitution of imports, driven by a single large private refining asset rather than trade policy. Automobiles, protected at a rate designed explicitly to reward local assembly, show 29.0% aggregate capacity utilization across 34 licensed assemblers and a policy framework that has never acquired the force of law. Rice, the most heavily protected sector, remains the single most-smuggled commodity through Nigerian customs checkpoints as of the first quarter of 2025. These findings extend Corden's (1971) effective-protection framework and Krueger's (1974) rent-seeking model to a contemporary African case, and they corroborate Gao et al.'s (2023) finding that decades of Nigerian agricultural protectionism have left import dependence largely unchanged. The paper also finds that Nigeria's AfCFTA commitments — gazetted only in 2025 — cannot be shown to constrain sector-level outcomes in this window for two of the three sectors studied, a limitation on the "AfCFTA versus protectionism" narrative common in policy commentary. The central implication is that tariff schedules are a weak proxy for industrial-policy seriousness; the presence of a credible, scaled domestic supply response is the better predictor of substitution outcomes.

Keywords: trade policy, tariff protection, industrial policy, Nigeria, AfCFTA, import substitution, effective rate of protection

1. Introduction

Nigeria's trade policy since the May 2023 inauguration of President Bola Tinubu has been characterized by rapid, sometimes contradictory action: an overnight removal of the petroleum subsidy, a currency float that depreciated the naira by over 40% in a single year, a temporary zero-duty window for imported food staples, and a 15% petroleum import duty that was approved and reversed within three weeks. Running beneath these unilateral moves is a slower, largely separate process — Nigeria's accession to the African Continental Free Trade Area (AfCFTA), whose tariff concession schedule the country did not formally gazette until 2025, six years after signing the framework agreement.

A substantial body of policy commentary treats Nigeria's tariff regime as caught in tension between AfCFTA-driven liberalization and domestic protectionist reflexes (see Section 2.3). This paper tests whether that framing survives contact with sector-level evidence. It does so through a comparative case-study design across three sectors that received materially different tariff treatment over the same 2023-2025 window: refined petroleum, automobiles, and rice. The central research question is narrow and empirically tractable: **does the level of tariff protection a sector receives predict whether that sector actually substitutes imports with domestic production?**

The paper's contribution is twofold. First, it provides a rare recent empirical comparison of tariff outcomes across sectors within a single developing-economy trade regime, at a moment (2023-2025) when Nigeria's policy environment was unusually volatile and therefore analytically informative — natural variation in policy stringency substitutes for the controlled variation a researcher cannot otherwise engineer. Second, it re-examines classical trade-policy theory (effective protection, rent-seeking, infant-industry protection) against this variation and finds that the theory's traditional skepticism of tariff-led industrialization is corroborated more by the *presence or absence of a credible domestic supply response* than by the tariff rate itself — a distinction with direct relevance to how effective protection theory should be applied empirically.

2. Literature Review

2.1 Effective protection and the limits of nominal tariff rates

The starting theoretical premise of this paper — that a nominal tariff rate is a poor guide to a policy's actual economic effect — is not new. Corden's (1971) foundational work on the effective rate of protection (ERP) demonstrated that the protection a tariff structure confers on value added in a sector depends on the tariff differential between outputs and inputs, not on the output tariff alone. Nigeria's own automotive tariff structure is, in fact, a textbook ERP design: a substantial tariff on fully-built vehicle imports paired with near-zero rates on the completely-knocked-down (CKD) and semi-knocked-down (SKD) kits used by licensed local assemblers, intended to raise the effective protection on domestic assembly value-added well above the nominal tariff on the finished good. Corden's framework predicts this structure *should* incentivize local assembly. Section 4.2 examines whether it has.

2.2 Rent-seeking and the political economy of quantitative and tariff restrictions

Krueger's (1974) analysis of rent-seeking in trade-restricted economies — developed initially from case studies of Turkey and India — argues that quota and license-based trade restrictions generate competition for the rents created by scarcity, often diverting resources toward securing import licenses or protected status rather than toward productive investment. While Nigeria's automotive and rice tariffs are ad valorem rather than quota-based, the underlying dynamic Krueger identifies — a policy-created rent whose capture does not require, and may not incentivize, productive scale-up — is directly testable against the Nigerian evidence: a licensing regime (NAIDP for automobiles; import waivers and license allocation for rice and petroleum) that confers preferential access without a correspondingly enforced production or investment obligation.

2.3 Industrial policy: the Rodrik critique and its conditions for success

Rodrik's (2004) *Industrial Policy for the Twenty-First Century* reframed the debate away from a binary "protection versus free trade" question toward the institutional conditions under which industrial policy succeeds: policies that combine incentives with accountability (embedded autonomy, sunset clauses, monitorable performance benchmarks) tend to outperform open-ended protection. This paper treats Rodrik's conditions as a diagnostic checklist against the Nigerian cases, since it offers a more falsifiable test than the traditional protectionist/liberal binary that dominates commentary on Nigeria's trade policy specifically (Section 2.4 below).

2.4 Sector-specific literature on Nigeria

Automobiles. Two recent peer-reviewed studies examine Nigeria's automotive policy directly. Ugwueze, Ezeibe, and Onuoha (2020), writing in the *Review of African Political Economy*, trace automobile policy since 1960 and argue that inconsistent implementation has reinforced the dominance of non-indigenous manufacturers, concluding that policy consistency — not policy existence — is the binding constraint on indigenous production and job creation. Agbo (2020), in the *Journal of Science and Technology Policy Management*, similarly finds that Nigeria's automotive policies are not self-sustaining and that the import-substitution approach omitted key components of a functioning automotive manufacturing base, leaving the sector "footloose" and exposed to currency shocks given Nigeria's oil-dependent exchange rate. Both papers predate the 2023-2025 window studied here but establish a documented pattern — persistent underperformance despite nominal protection — that this paper's primary NADDC capacity data (Section 4.2) allows us to quantify directly for the first time at a 34-firm, economy-wide level.

Rice and agriculture. Gao, Zhao, Xu, and Ndidiamaka (2023), in *Frontiers in Sustainable Food Systems*, map Nigeria's rice self-sufficiency policy narrative since the 1970s and conclude that "seesawed trade policy... leaves imported rice dependence unchanged" despite five decades of protectionist intervention. Bouët, Laborde, Traoré, and Malhotra (2019), writing for the International Food Policy Research Institute (IFPRI), document Nigeria's 2019 land-border closure and argue explicitly that unilateral import prohibitions of this kind "go against both the letter and the spirit" of regional trade integration commitments, naming rice and used automobiles among the affected product categories — an early documented instance of the same tension this paper examines in the 2023-2025 window under AfCFTA specifically.

AfCFTA and the openness-protectionism tension. Boysen (2024), in *The World Economy*, analyzes submitted AfCFTA tariff offers across signatories and finds "significant data gaps and inconsistencies" between AfCFTA's liberalization modalities and members' actual tariff classification practice; Nigeria specifically is characterized as committing to liberalize 90% of tariff lines continent-wide while "remain[ing] largely protectionist in agriculture, shielding staples like rice, sugar, and poultry behind high tariffs and import controls, citing food security." The World Trade Organization's November 2024 Trade Policy Review of Nigeria reaches a compatible conclusion through a different lens, with member states noting that Nigeria's long-standing agricultural import prohibitions and tariff peaks carry a risk of contributing to food insecurity and higher food prices — the stated opposite of AfCFTA's integration goals.

A literature gap: refined petroleum. No peer-reviewed academic literature was located examining Nigeria's 2023-2025 petroleum subsidy removal, downstream deregulation, or the Dangote Refinery's entry into domestic PMS production — the most consequential development in this paper's petroleum case study. This is unsurprising given the recency of these events (the refinery began petrol production in September 2024) but is itself a finding worth noting: the academic literature on Nigerian trade and industrial policy has not yet caught up to the most significant real-world substitution event covered in this paper. This paper's petroleum case study therefore relies on primary data (NBS trade statistics, NNPC/NMDPRA disclosures) and financial/trade press reporting rather than peer-reviewed secondary literature, a limitation discussed further in Section 6.

2.5 Positioning this paper

The reviewed literature converges on a shared skepticism of Nigerian tariff protection's effectiveness (Ugwueze et al., 2020; Agbo, 2020; Gao et al., 2023) and a shared observation that Nigeria's trade policy is more protectionist in practice than its regional commitments (AfCFTA) suggest in principle (Boysen, 2024; Bouët et al., 2019; WTO, 2024). What the existing literature has not done is compare sectors with *substantially different* tariff treatment within the same short policy window to isolate what — if not the tariff rate — actually predicts substitution outcomes. That comparison is this paper's central empirical contribution.

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3. Data and Methodology

This paper uses a structured comparative case-study design across three sectors — refined petroleum, automobiles, and rice/agriculture — selected because each was subject to distinct, well-documented tariff treatment during the 2023-2025 window and because at least one primary or near-primary data source was available for each. Data were compiled from national statistical releases (Nigeria's National Bureau of Statistics foreign trade statistics; USDA Foreign Agricultural Service grain and feed reporting), a primary sectoral regulator document (the National Automotive Design and Development Council's April 2024 capacity utilization table, covering all 34 then-licensed assemblers), and contemporaneous financial and trade press reporting where government data were unavailable at the needed frequency (e.g., the timeline of the October-November 2025 petroleum duty episode). Every figure is traced to a named, dated source, catalogued in `data/raw/{petroleum,automobiles,rice,afcfta}/sources.md`; where sources conflicted, both figures are reported rather than reconciled silently.

Three limitations bear directly on the inferences this paper can support. First, Nigeria's AfCFTA Provisional Schedule of Tariff Concessions was gazetted only in 2025, and no source — primary or secondary — was located specifying where refined petroleum or automobiles sit within that schedule; only rice's placement on an exclusion or long-timeline "sensitive" list is independently corroborated. This asymmetry means the AfCFTA-specific claims in Section 5.3 apply confidently to rice alone. Second, most Nigerian trade-value data available in secondary reporting are naira-denominated and not volume-adjusted; 2024 in particular saw a 40.9% naira depreciation that mechanically inflates naira-value figures independent of underlying import volumes, and this is flagged wherever it materially affects interpretation. Third, one figure from the underlying Phase 1 data-collection pass required correction during analysis: the NADDC capacity table's aggregate utilization was originally miscalculated at approximately 24%; recomputing directly from the table's own rows yields 29.0% (93,950 actual units against 323,650 installed units annually), which this paper uses throughout.

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4. Findings

4.1 Refined petroleum: substitution without tariff protection

Nigeria operated without a dedicated import tariff on petrol (PMS) or diesel (AGO) for nearly all of 2023-2025; the only broadly applicable charge was a general 0.5% levy on non-African imports introduced by the Finance Act 2023, not a petroleum-specific instrument. The one dedicated petroleum tariff action of the period — a 15% ad valorem duty approved on 21 October 2025 and explicitly framed by officials as protecting the newly operational Dangote Refinery — was suspended on 13 November 2025, before its 30-day transition window had even elapsed, following opposition from marketers and civil society over projected pump-price increases. The duty was never collected.

Substitution nonetheless occurred, and quickly. The Dangote Refinery (650,000 barrels-per-day nameplate capacity) began PMS production in September 2024 and reached an estimated 85% utilization by mid-2025. Nigeria's petrol import bill fell from a currency-inflated 2024 peak of N15.42 trillion to N8.96 trillion in 2025 — a 42% decline — and petrol's share of total Nigerian imports fell from 13.64% in the first quarter of 2025 toward near-zero within a year, coinciding with NMDPRA's move to halt new PMS import licenses on the stated grounds that domestic production was meeting demand. The mechanism sustaining this substitution — a naira-denominated crude supply arrangement between NNPC and Dangote — allocated 82 million barrels between October 2024 and October 2025 (60% naira-settled), but has since shown institutional strain: a reported supply shortfall of approximately 79.5 million barrels between late 2025 and mid-2026, and an active legal dispute over import-license enforcement. This sector's substitution is real but should be read as institutionally immature rather than settled.

4.2 Automobiles: effective protection without the predicted response

Nigeria's automotive tariff structure closely follows the effective-protection logic Corden (1971) describes: a substantial combined duty and levy on fully-built vehicle imports (35% used, 40% new) against a 0% rate for CKD kits and 10% for SKD kits used by licensed assemblers — a structure designed explicitly to make local assembly value-added more profitable than importing finished vehicles. The National Automotive Industry Development Plan (NAIDP), first introduced in 2013 at an even steeper differential (70% combined on finished imports), is the policy instrument underlying this structure.

A primary NADDC document dated April 2024, covering all 34 then-licensed assemblers, allows this paper to test Corden's predicted response directly. Summed across the full table: installed annual capacity of 323,650 units against actual output of 93,950 units, an aggregate utilization of 29.0%. The distribution is as informative as the aggregate: Innoson (the only wholly Nigerian-owned mass-market original equipment manufacturer in the list) operates at 70% utilization, while Mikano International — with the largest single installed capacity in the table at 46,000 units annually — operates at 6.5%. Twenty of 58 companies historically licensed under NAIDP had suspended operations entirely by March 2025, representing an estimated $89.6 million in stranded investment.

This pattern is consistent with Krueger's (1974) rent-seeking framework: a policy that confers a substantial tariff concession on license-holders without a binding, monitored production obligation creates conditions in which some firms may capture the concession without scaling production accordingly. It is also consistent with Rodrik's (2004) diagnostic that industrial policy without embedded accountability underperforms — the NAIDP framework has never acquired the force of law; its original ten-year statutory term lapsed in 2024, and as of October 2025 industry stakeholders, including NADDC's own Director-General, were still publicly demanding legislative passage of a "NAIDP Bill." As one representative quote from that reporting states: *"Nobody wants to commit serious capital to the auto industry without adequate laws to protect their investments."* This corroborates Ugwueze et al.'s (2020) and Agbo's (2020) shared conclusion that policy inconsistency, not the absence of protection, is the binding constraint on Nigerian automotive development — this paper's contribution is a economy-wide utilization figure (29.0%) that quantifies the scale of that underperformance directly.

4.3 Rice: the heaviest protection, the most persistent leakage

Rice carries the highest effective tariff of the three sectors studied: a combined 70% duty and levy, reduced to 30% for integrated millers able to demonstrate backward integration into domestic paddy sourcing, a structure that has held roughly constant since at least 2013. This tariff sits atop Nigeria's 2019 land-border closure with Benin, Cameroon, Chad, and Niger, undertaken explicitly to curb rice and poultry smuggling; the border was only ever partially reopened in December 2020, with rice remaining formally banned even after general trade resumed.

The evidence that this combination has not achieved its stated goal is direct and recent: in the first quarter of 2025 alone, Nigerian Customs recorded 159 rice-smuggling seizure cases totaling 135,474 bags worth N939 million, making rice the single most-smuggled commodity that quarter — ahead of both refined petroleum products and narcotics. USDA Foreign Agricultural Service estimates place Nigeria's rice self-sufficiency at roughly 70-76% across 2022-2025, a ratio that appears to be improving mainly because forecast domestic production is *declining* (attributed to insecurity in growing regions and reduced mechanization investment) alongside a larger fall in import demand, plausibly a currency-driven affordability effect rather than a supply-side improvement — a pattern structurally similar to the demand effects observed in the petroleum and automobile cases. This finding directly corroborates Gao et al.'s (2023) conclusion that Nigeria's decades-long protectionist rice policy has left underlying import dependence largely unchanged despite repeated escalation of the protective instrument.

4.4 Cross-sector synthesis

Table 1 summarizes the three cases against their effective tariff level and outcome.

Table 1. Tariff level and substitution outcome by sector, Nigeria, 2023-2025

Sector	Effective tariff (2025)	Primary policy instrument	Substitution outcome
Refined petroleum	≈0.5%	Signaling (an approved-then-suspended duty; import-license administration)	Fastest and largest substitution of the three cases, driven by a single private capital asset
Automobiles	35-40% (0-10% for licensed-assembler kits)	Sustained tariff differential under NAIDP, without statutory force	29.0% aggregate capacity utilization; substitution far below what the tariff structure was designed to produce
Rice/agriculture	70%	Sustained high tariff plus intermittent border closures and duty waivers	Persistent large-scale smuggling; import dependence largely unchanged over decades

The ordering in Table 1 is the inverse of what a simple tariff-as-protection thesis would predict: the least-protected sector achieved the most substitution, and the most-protected sector shows the weakest enforcement outcome. This is consistent with Corden's (1971) original caution that nominal or even effective tariff rates describe policy *intent*, not realized economic outcome, and with Rodrik's (2004) argument that the presence of accountability mechanisms — not the size of the incentive — determines whether industrial policy succeeds. The variable that best explains the divergence across these three cases is not the tariff line but whether a credible, adequately capitalized domestic supply response existed independent of the tariff: a single large refining asset in petroleum; a fragmented base of 34 individually under-scaled assemblers in automobiles; and a still-fragmented milling sector, alongside porous enforcement, in rice.

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5. Discussion

5.1 Effective protection theory revisited

Corden's (1971) framework predicts that a well-designed tariff differential should raise effective protection on domestic value-added and thereby incentivize local production. Nigeria's automotive tariff structure satisfies the formal conditions of that prediction, yet the realized outcome (29.0% utilization) falls far short of it. This is not necessarily a refutation of ERP theory — Corden's own later work acknowledged that effective protection is a necessary, not sufficient, condition for a supply response — but it is a clear illustration that the theory's predictive power collapses without complementary institutional conditions (Rodrik, 2004) and in the presence of rent-seeking dynamics (Krueger, 1974) that this paper's evidence — the wide utilization spread across otherwise similarly-tariffed firms, and NAIDP's persistent lack of statutory force — is consistent with, though this paper cannot fully adjudicate between rent-seeking and simple state capacity failure as the dominant explanation without firm-level investment data beyond this paper's scope.

5.2 Why petroleum breaks the pattern

The petroleum case is the paper's most theoretically interesting because it substitutes imports fastest with essentially no tariff support, which on its face appears to contradict infant-industry and effective-protection reasoning entirely. The better reading is not that tariffs are irrelevant, but that Dangote's refinery investment was large enough, and sufficiently exogenous to Nigerian tariff policy specifically (financed against global capital markets and crude-supply arrangements, not against a promised tariff wall), that it did not require tariff protection to reach viable scale. This suggests a boundary condition on effective-protection theory worth stating explicitly: ERP-style tariff design matters most for inducing supply responses in sectors where domestic capital formation is itself constrained by scale economies that no individual protected firm can reach alone (as in the fragmented automotive and rice-milling sectors) — and matters least where a single actor can internalize the full scale economics of the sector independent of trade policy.

5.3 The limits of the AfCFTA framing

A significant portion of Nigerian trade-policy commentary — including Boysen (2024) and the WTO's own 2024 review — frames Nigerian protectionism against the country's AfCFTA liberalization commitments. This paper's evidence supports that framing for rice specifically, where an exclusion-list classification is independently corroborated by multiple sources, but not for petroleum or automobiles, where no source (primary or secondary) was located specifying an AfCFTA tariff category at all. Given that Nigeria only gazetted its AfCFTA schedule in 2025, the more defensible claim over this specific window is that Nigeria's *unilateral* tariff and non-tariff tools — subsidy policy, import-license administration, ad hoc duty waivers — did far more to shape sector outcomes than AfCFTA commitments did. Whether AfCFTA becomes a binding constraint on Nigerian sector policy is an empirical question for a later research window, not one this paper's 2023-2025 data can answer.

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6. Conclusion and Limitations

This paper compared three Nigerian sectors under materially different tariff regimes over 2023-2025 and found that effective tariff level does not predict import-substitution outcomes; the presence of a credible, adequately scaled domestic supply response does. This finding extends established skepticism of tariff-led industrialization (Corden, 1971; Krueger, 1974; Rodrik, 2004) with a contemporary, cross-sector empirical test, and it corroborates existing Nigeria-specific findings on automotive policy inconsistency (Ugwueze et al., 2020; Agbo, 2020) and agricultural protectionism's limited effectiveness (Gao et al., 2023) using newly available primary data — most notably a corrected, economy-wide capacity-utilization figure for Nigeria's licensed automotive sector.

The paper's limitations should constrain how its findings are generalized. First, the three-sector comparison, while useful for isolating the tariff-versus-supply-response distinction, is not a random or representative sample of Nigerian tariff lines, and the mechanism identified (single large asset vs. fragmented sector) may not generalize cleanly to sectors with different capital-intensity or scale-economy profiles. Second, the absence of peer-reviewed literature on the petroleum case (Section 2.4) means that case rests more heavily on primary statistical data and trade press than the other two, a gap the literature has not yet had time to fill given the recency of events. Third, this paper's AfCFTA findings are necessarily provisional given that Nigeria's own tariff schedule under the agreement was gazetted only in 2025; a follow-up study conducted after 2027-2028, once AfCFTA implementation has matured, would be better positioned to test whether Nigeria's regional commitments have begun to bind on sector-level policy in the way current commentary anticipates.

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Primary data sources (Nigeria National Bureau of Statistics, NADDC, USDA Foreign Agricultural Service, NNPC/NMDPRA disclosures) and full trade-press citations underlying Section 4 are catalogued in `data/raw/{petroleum,automobiles,rice,afcfta}/sources.md`, with structured datasets in `data/processed/*.csv` and analysis code in `models/*_analysis.py`.